Composite functions

1 Differentiate the following:

a $y = e^x \sin x$

b $y = e^{\sin x}$

c $y = e^x \cos 3x$

d $y = e^{3x} \cos\left(\frac{x}{3}\right)$

e $y = e^{-x} \sin 4x$

f $y = \cos(\ln x)$

g $y = \cos(\ln(-5x))$

h $y = \cos x \ln x$

i $y = e^x \ln x$

j $y = e^{2x} \ln(4x)$

k $y = \log_e(\ln x)$

l $y = \dfrac{\ln x}{e^{2x}}$

m $y = e^{-2} \ln\left(-6 + x^{-3}\right)$

n $y = \cos\left(\frac{\pi t}{2} - \frac{\pi}{3}\right)$

o $y = x^2 \sin\left(\frac{1}{x}\right)$

p $y = 4\sin\left(\frac{x}{5}\right) - 6e^{2x} + x^{-8}$

q $y = \ln\left(x^2 - 7x - 12\right) - \sqrt{11x}$

r $y = \left(e^{-5x^2} + \cos x\right)^5$

s $y = \dfrac{e^{-0.2x}}{\sin\left(\frac{\pi}{4}x\right) - x^4}$

t $y = (\cos x + \sin x)\, e^{6x}$

u $y = \dfrac{\cos\left(4x - \frac{10\pi}{11}\right)}{(x+3)^2}$

2 Find the derivative of the following:.

a $e^{5x}\cos(x)$

b $x^3 \sin x$

c $\cos^2(x)$

d $\dfrac{\cos x - \sin x}{\cos x + \sin x}$

e $\dfrac{4x}{\sin x}$

f $e^{3x}\cos\left(5x + \frac{4\pi}{7}\right)$

g $\dfrac{\ln x}{\sin x}$

3 For each of the following curves and given points:

i Find an expression for $\frac{dy}{dx}$.

ii Find the exact value of the gradient of the curve at the given point.

a $y = \cos 3x$ at $x = \frac{\pi}{18}$.

b $y = 6 - \cos 3x$ at $x = \frac{\pi}{9}$.

c $y = \sin 4x$ at $x = \frac{\pi}{16}$.

d $y = \sin^2(4x)$ at $x = \frac{\pi}{32}$.

e $y = \cos^2(2x)$ at $x = \frac{\pi}{24}$.

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4 Consider the expression $e^{\cos x}\sin\left(e^x\right)$.

a If $u = e^{\cos x}$, find $\dfrac{du}{dx}$.

b If $v = \sin\left(e^x\right)$, find $\dfrac{dv}{dx}$.

c Hence, find the derivative of $y = e^{\cos x}\sin\left(e^x\right)$.

5 Consider the function $y = \dfrac{4x^2 + e^x}{\cos 7x}$.

a If $u = 4x^2 + e^x$, find u'.

b If $v = \cos 7x$, find v'.

c Hence, find y'.

6 Consider the function $y = xe^x$.

a Show that $e^{x+\ln x} = xe^x$.

b Hence, find $\dfrac{dy}{dx}$, without using the product rule.

7 Determine $\dfrac{dy}{dx}$, given that $y = u^5$ and $u = \ln(x+5)$.

8 Consider the expression $e^x\left(1+\ln 0.3x\right)^{-2}$.

a If $y = \left(1+\ln\left(0.3x\right)\right)^{-2}$, find $\dfrac{dy}{dx}$.

b Hence, find the derivative of $y = e^x\left(1+\ln 0.3x\right)^{-2}$.

9 Consider the equation $y = \left(2\ln\left(-3x-x^2\right) - \cos\left(\dfrac{\pi}{3}x\right) + \dfrac{1}{x^4}\right)^3$.

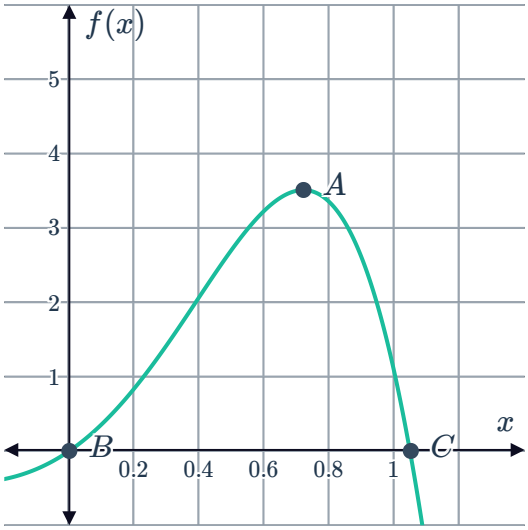
a Find the derivative of $2\ln\left(-3x-x^2\right)$.

b Hence differentiate $y = \left(2\ln\left(-3x-x^2\right) - \cos\left(\dfrac{\pi}{3}x\right) + \dfrac{1}{x^4}\right)^3$.

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Stationary points

10 The graph of the function $f(x) = e^{2x} \sin 3x$ is shown:



a Find $f'(x)$.

b Find the x-intercepts, B and C.

c Determine the coordinates of point A, correct to two decimal places.

11 Consider the function $f(x) = x^3 \ln x$, over the domain $e^{-\frac{1}{2}} \leq x \leq e^{-\frac{1}{4}}$.

a Find an expression for $f'(x)$.

b Find the exact values of x such that $f'(x) = 0$.

c Complete the table of values:

x	$e^{-\frac{1}{2}}$	$e^{-\frac{1}{3}}$	$e^{-\frac{1}{4}}$
$f(x)$			

d Hence, state the nature and location of any stationary points.

e Calculate the exact global minimum over the domain.

f Calculate the exact global maximum over the domain.

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12 Consider the function $y = \dfrac{\ln 3x}{e^{3x}}$.

a Find the x-intercept of the function.

b Find an expression for $\dfrac{dy}{dx}$.

c Show that the turning point of the graph occurs when $3x \ln 3x - 1 = 0$.

d An approximation to the solution of the equation $3x \ln 3x - 1 = 0$ is $x = 0.58$. Complete the table of values, correct to two decimal places.

x	0.01	0.58	1.58
$\dfrac{dy}{dx}$		0	

e Hence, state the nature of the turning point at $x = 0.58$.

f State the coordinates of the turning point, correct to two decimal places.

13 Determine the values of the non-zero constants, a and b, for the following function, given it has a turning point at $(0.25, 1)$:

$$f(x) = axe^{bx}$$

14 Consider the function $f(x) = \ln(\sin 2x)$.

a State the values of x between 0 and 2π for which the function is defined.

b Determine the values of x in the domain $0 \leq x \leq 2\pi$, for which the function has a maximum value.

c State the maximum value of the function.

Applications

15 The curve $y = x \cos x$ passes through the point Q, $\left(\dfrac{\pi}{2}, 0\right)$.
Find the equation of the tangent at point Q.

16 Find the equation of the tangent to the following curves:

a $y = e^x - 3 \sin x$ at $x = \dfrac{3\pi}{2}$.

b $y = e^{\cos x}$ at $x = \dfrac{3\pi}{2}$.

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17 Researchers have created a model to project the country's population for the next 10 years, where P is the population (in thousands), t years from now. The model is defined by the function:

$$P(t) = \frac{57\,460e^{\frac{t}{7}}}{t+13}$$

a State the current population of the country.

b According to the model, state the current rate of growth of the population, to the nearest thousand.

c Find the rate of population growth 7 months from now, to the nearest thousand.

d Find the rate of population growth 10 years from now, to the nearest thousand.

18 The displacement of a particle moving in rectilinear motion is given by:

$$x(t) = (t-2)^2 + \sin(t-4\pi) + 3t + 25$$

a State the initial displacement of the particle.

b Write an expression for the velocity of the particle, $v(t) = x'(t)$.

c Write an expression for the acceleration of the particle, $a(t) = v'(t)$.

19 A charged particle moves back and forth about the fixed point $x = 0$ (called the origin). Its position, x cm from the origin, after t seconds is given by the equation:

$$x = \sin\left(\pi e^{2t}\right)$$

a Find the particle's initial position.

b Find an expression for the velocity of the particle , $v(t) = x'(t)$.

c Find the exact times for the first and second occasion that the particle comes to a stop.

d Describe the position of the particle when it first comes to a stop, $v(t) = 0$.

e Describe the position of the particle when it comes to a stop for the second time.

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